

SANTIAGO RAMÓN PINEDO PEÑA

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PROFESSIONAL SUMMARY

Hybrid **Data Platform & Quality Engineer (SDET)** with 4+ years of experience bridging scalable big data architectures and zero-defect automated validation frameworks. Specialized in high-concurrency ETL/ELT pipelines (PySpark, Scala) within AWS ecosystems and advanced UI/HMI automated testing via Computer Vision and OCR. Proven track record taking total ownership of data flow integrity for Tier-1 global projects (**BBVA, Ford Motor Co.**). Brings an academic foundation in Human-Machine Interaction and Machine Learning into robust, audit-ready platform engineering.

TECHNICAL SKILLS

- **Data Engineering & Cloud:** PySpark, Apache Spark, Scala, AWS (S3, Athena, CloudWatch, SageMaker), Control-M, ETL/ELT Pipelines, Data Mesh, Data Governance, Looke Studio, Oracle Analytics.
- **Programming & DBs:** Python, Java, Scala, SQL (PostgreSQL, MySQL), Oracle XE DB, Go, JavaScript, CSS.
- **AI, Vision & Automation:** OpenCV, OCR (Optical Character Recognition), Squish, Slash Framework, CAN Protocol, Automotive ECUs, Machine Learning, Neural Networks, Deep Learning, Transformers, LSTMs.
- **DevOps, Workflow & UI:** Git, GitHub, Bitbucket, Jira, Confluence, Figma, Balsamiq Mockups.

WORK EXPERIENCE

Data Platform Engineer — Core Banking (BBVA Project)

Bluetab, an IBM Company | July 2025 – Present

- **Big Data Architecture:** Architected and deployed fault-tolerant PySpark and Scala ETL/ELT pipelines for the bank's Advanced Data Architecture (ADA), centralizing massive financial datasets into **AWS S3**.
- **Orchestration & Querying:** Designed complex execution meshes and automated batch jobs using **Control-M** and ADA engines; leveraged **AWS Athena** to perform high-performance serverless queries over multi-gigabyte transactional tables.
- **Data Quality & Monitoring:** Enforced enterprise Data Governance by programming automated data-cleansing rules and configuring **AWS CloudWatch** to monitor pipeline health, preventing corruption in downstream analytical models (including **AWS SageMaker** inputs).
- **Agile Delivery:** Managed technical debt, version control, and asynchronous collaboration using **Jira, Confluence, and Bitbucket/GitHub** within a high-velocity financial squad.

SDET & AI Automation Engineer (Ford Motor Co. Project)

Estrateg-IA | April 2024 – July 2025

- **Computer Vision Validation:** Programmed custom **OpenCV** and **OCR** algorithms in Python to automate the visual detection of dynamic icons, text, and rendering layers on vehicle display screens.
- **Hardware-in-the-Loop (HIL):** Engineered 150+ automated end-to-end test scripts using **Squish**, the **Slash** framework, and the **CAN protocol** to validate vehicle Electronic Control Units (**ECUs**), the **HMI**, and Ford **SYNC** flows.
- **Defect Prevention:** Intercepted an average of 3 critical background logic and UI rendering defects per month, significantly reducing regression bottlenecks during firmware release cycles.
- **Global Synchronization:** Operated 100% in English, maintaining version control repositories (**GitHub**) and documenting test execution matrices in Jira for international engineering teams.

Lead Software & UX Analyst

Exhos Technology | July 2023 – April 2024

- **System Blueprinting:** Led technical discovery, algorithmic logic design, and structural flow architecture for internal digital products.
- **Interactive Prototyping:** Translated complex business requirements into high-fidelity technical mockups using **Figma** and **Balsamiq Mockups**.
- **Quality Gatekeeping:** Acted as the bridge between UX and Engineering, executing functional validation of the developers' code against the algorithmic specifications documented in Jira.

EDUCATION & CERTIFICATIONS

- **B.S. in Computer Engineering** (Specialization in Human-Machine Interaction)
Universidad Autónoma del Estado de México (2017 – 2023)
Relevant Coursework: Machine Learning, Computer Vision, Robotics, Genetic Algorithms, Multimedia Systems.
- **Certified Six Sigma Green Belt** — Estrateg-IA
- **Google Data Analytics Professional Certificate** — Coursera / Google
- **Python and Data Science in OCI** — Oracle / Alura

LANGUAGES

- **Spanish:** Native
- **English:** Professional Working Proficiency (Full professional fluency in technical documentation and global team syncs).
- **Japanese:** A1 (Active study).